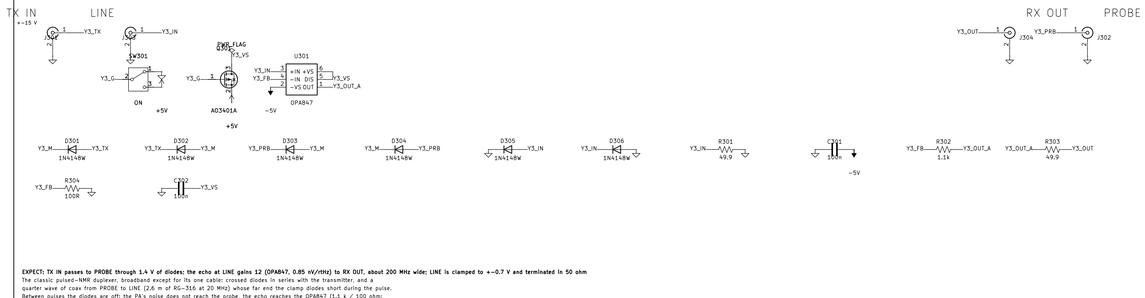
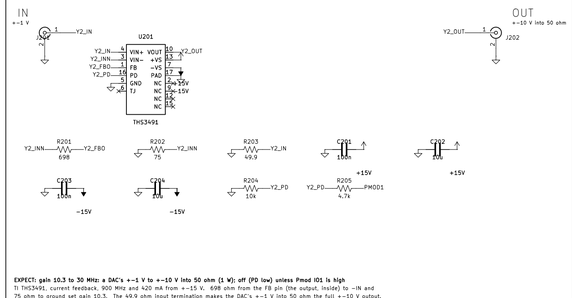
Y3 duplexer and preamp, OPA847

cells (0.0), (1.0)



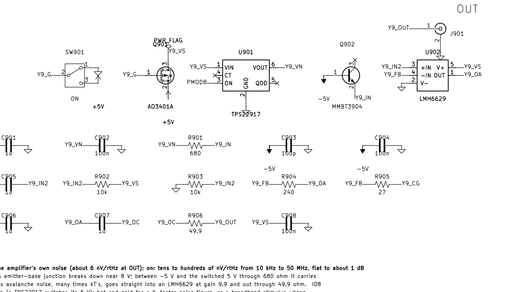
Y2 pulse amplifier, THS3491

cell (2.0)



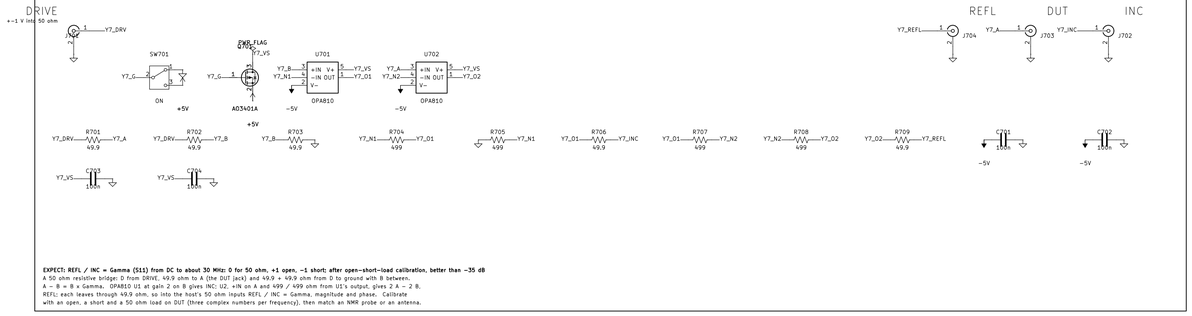
Y9 noise source, MMBT3904 + LMH6629

cell (3.0)



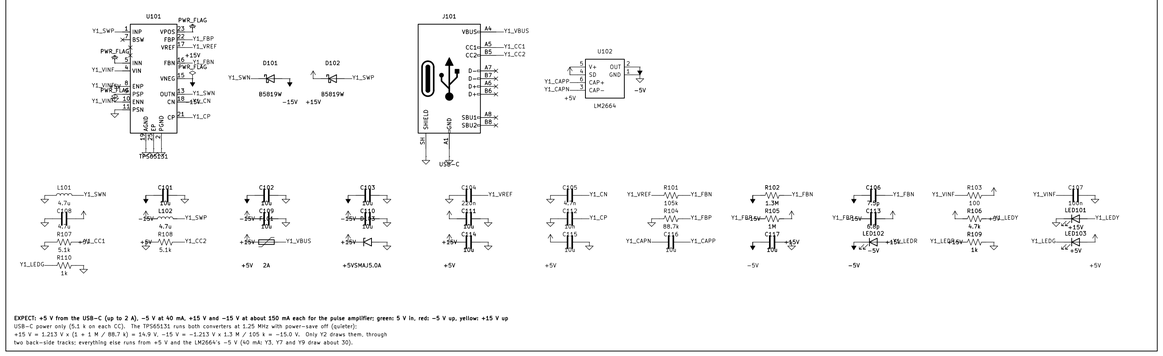
Y7 reflection bridge, 2 x OPA810

cells (0.1), (1.1)



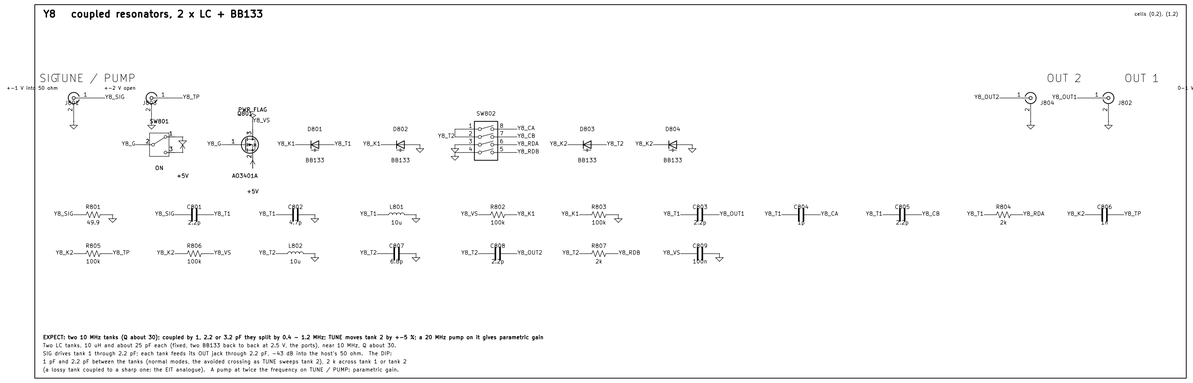
Y1 power, USB-C 5 V in: +5, −5, +15, −15 out

cells (2.1), (3.1)



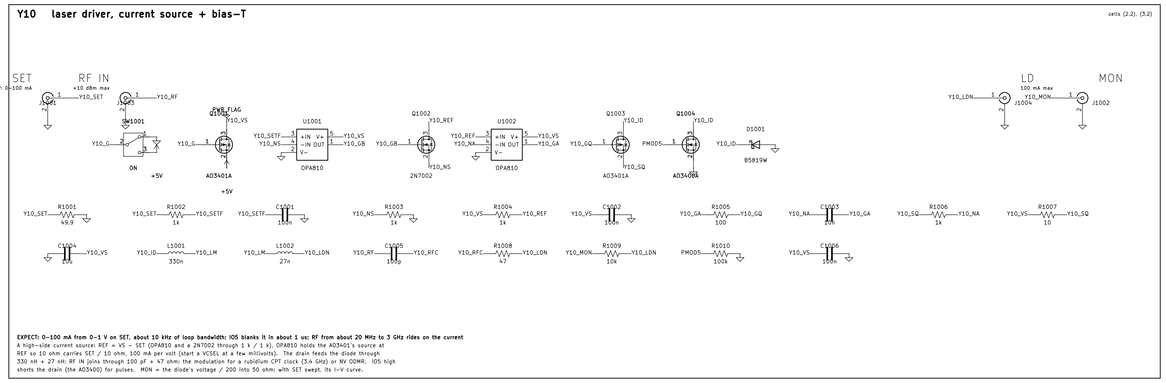
Y8 coupled resonators, 2 x LC + BB133

cells (0.2), (1.2)



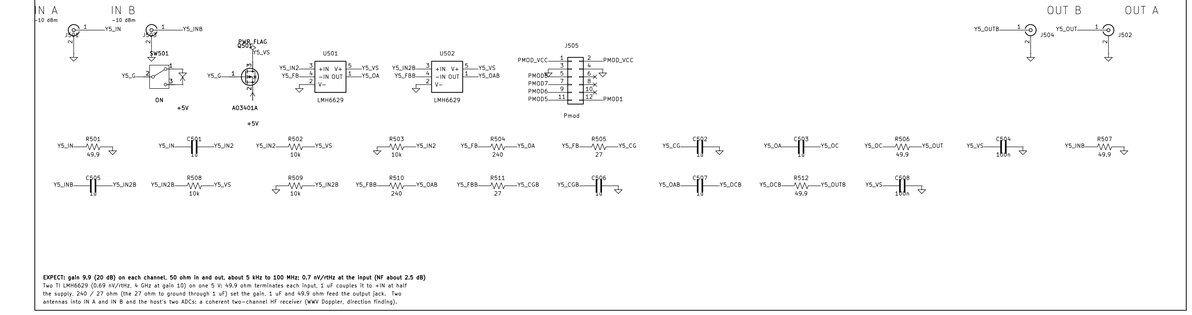
Y10 laser driver, current source + bias-T

cells (2.2), (3.2)



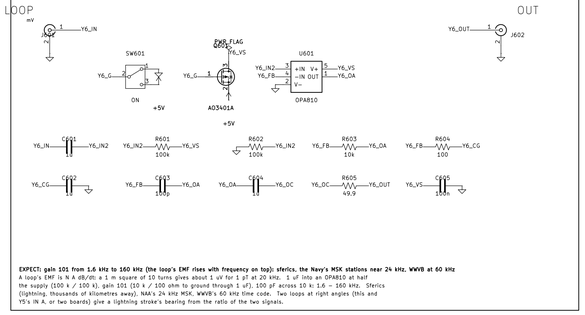
Y5 HF receiver, 2 x LMH6629

cells (0.3), (1.3)



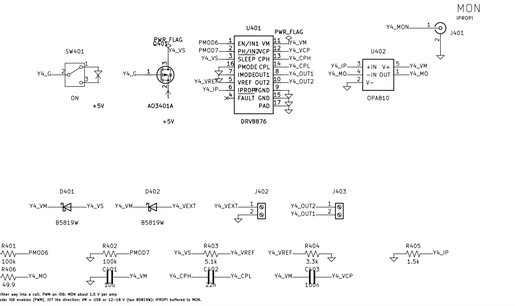
Y6 VLF loop amplifier, OPA810 x101

cell (2.3)



Y4 coil driver, DRV8876

cell (3.3)



GND GND
no mounting holes: four stick-on rubber feet, marked on the back

PWR_FLAG
ground loops for the scope's clips, one per channel, in the board's margins

Drawn by scripts/gen_sch.py from boards/phmod.py; the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
a pulse amplifier and duplexer, an HF receiver, VLF loops, a bridge, coupled resonators, a coil driver
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/

Sheet: / File: phmod.kicad_sch		
Title: PhMOD: magnetic resonance, the ionosphere and light		
Size: A1	Date: 2026-10-25	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1